Chapter Four

Utility

Utility

- Central concept in economics.
- Idea is to measure happiness or satisfaction
- Traced back to utilitarianism
- Neo-classical interpretation

Preferences - A Reminder

- $x \succ y$: x is strictly preferred to y.
- $x \sim y$: x and y are equally preferred.
- $x \succsim y$: x is preferred to y.

Utility Functions

- **Utility functions are numerical functions by which we represent preferences.**

- **A function U(x) represents a preference relation $\succsim$ if and only if:**

$$x' \succsim x'' \iff U(x') >= U(x'')$$

Utility Function

- **A preference relation that is complete, reflexive, transitive and continuous can be represented by a continuous utility function.**
- **Continuity means that small changes in the bundles lead to small changes in the preference level.**

Utility Functions

- **Utility is an ordinal (i.e. ordering) concept.**
- ***Example:*** **if U(x) = 6 and U(y) = 2 then bundle x is strictly preferred to bundle y. But x is not preferred three times as much as is y.**

Utility Functions & Indiff. Curves

- **Consider the bundles (4,1), (2,3) and (2,2).**
- **Suppose (2,3) $\succ$ (4,1) $\sim$ (2,2).**
- **Say U(2,3) = 6 > U(4,1) = U(2,2) = 4.**
- **Call these numbers utility levels.**

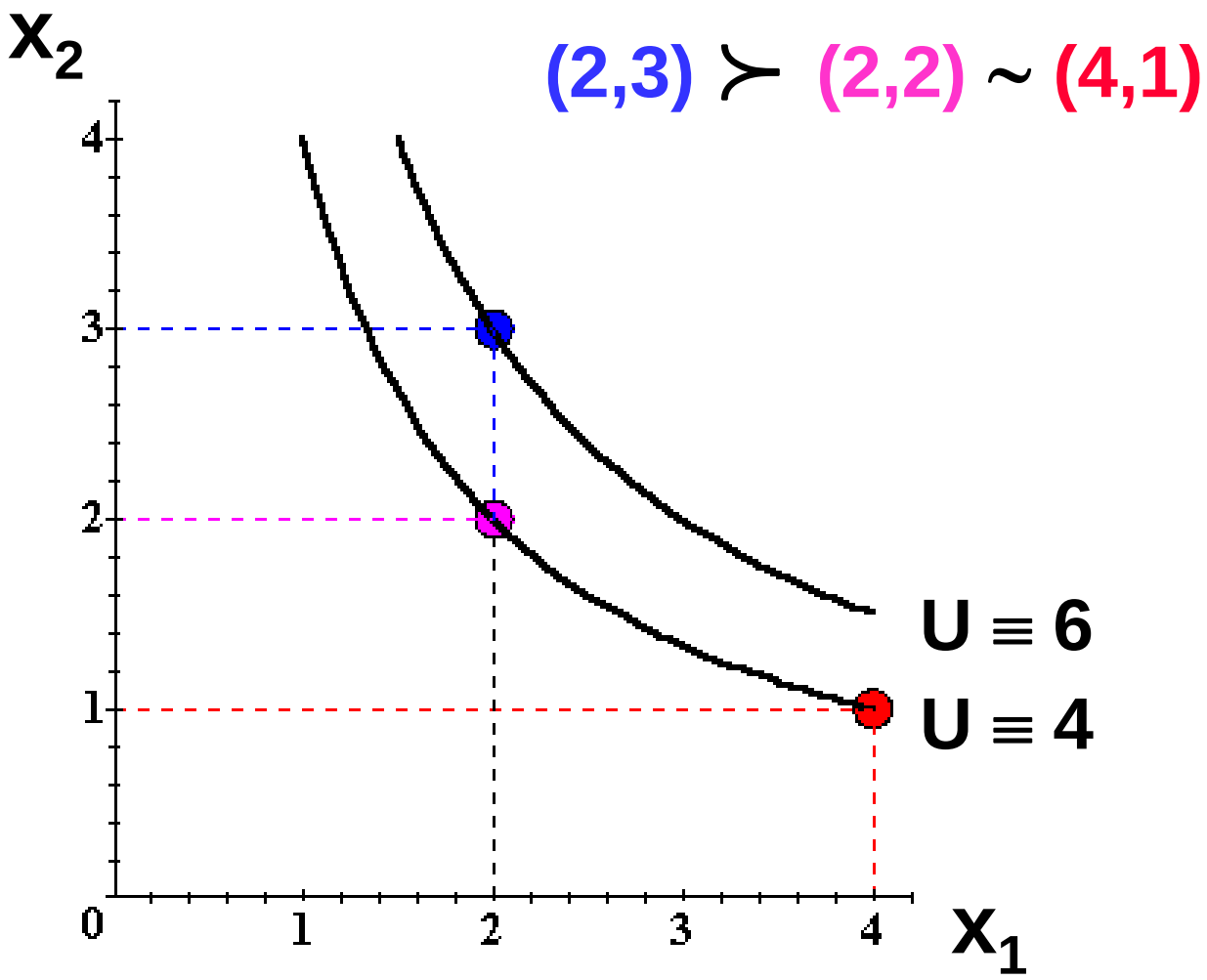
Utility Functions & Indiff. Curves

- **An indifference curve contains all equally preferred bundles.**
- **Therefore, an indifference curve contains all the bundles that have the same utility level.**

Utility Functions & Indiff. Curves

- **So the bundles (4,1) and (2,2) are in the indiff. curve with utility level U $\equiv$ 4**
- **But the bundle (2,3) is in the indiff. curve with utility level U $\equiv$ 6.**
- **On an indifference curve diagram, it looks as follows:**

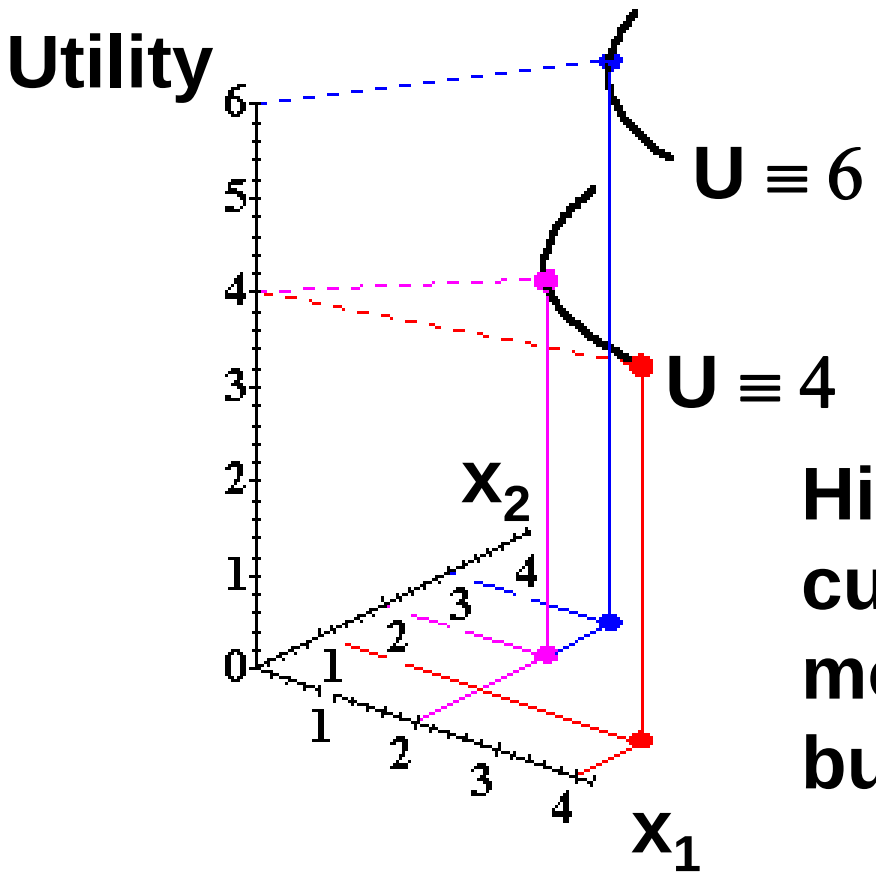
Utility Functions & Indiff. Curves



Utility Functions & Indiff. Curves

- **Another way to visualize this same information is to plot the utility level on a vertical axis.**
- **3D visualization of preferences**

Utility Functions & Indiff. Curves

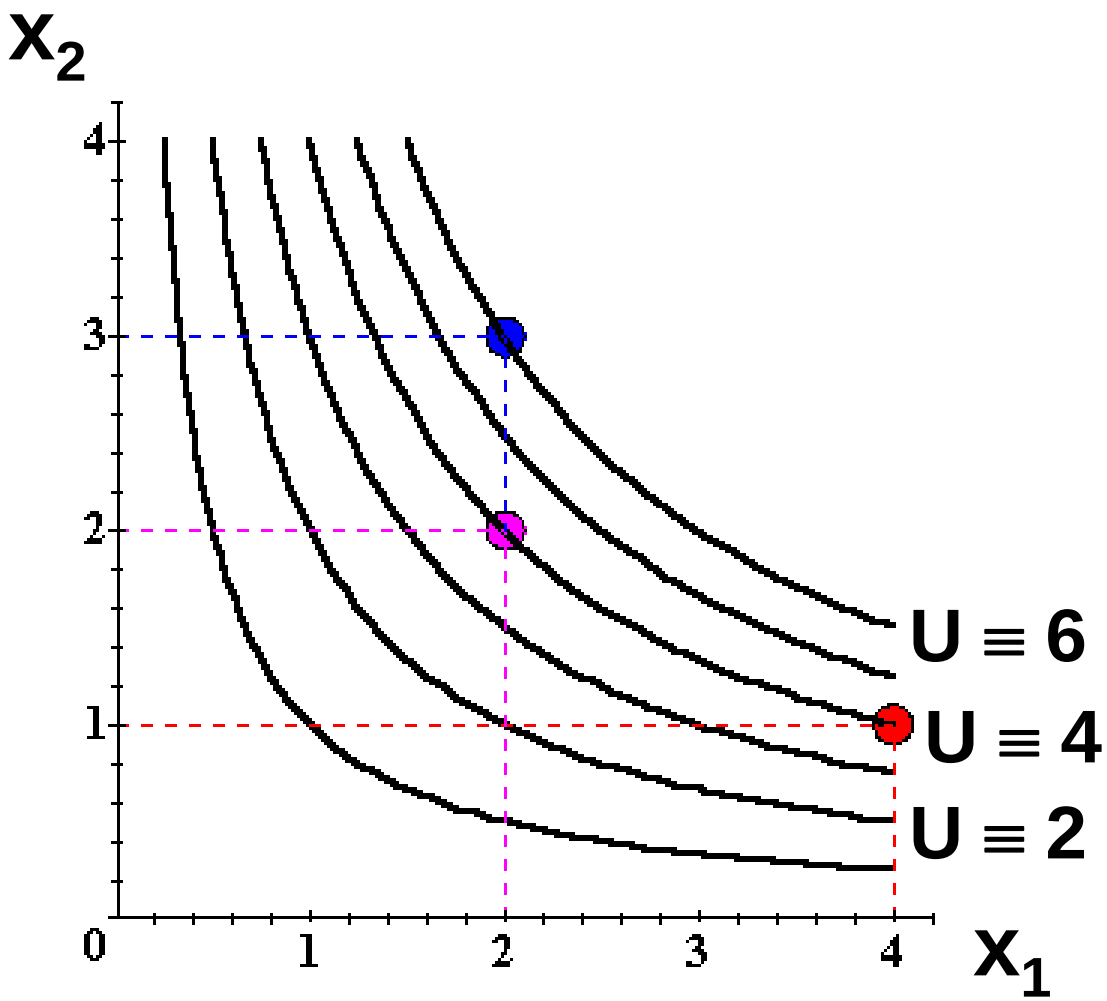


Higher indifference curves contain more preferred bundles.

Utility Functions & Indiff. Curves

- **Comparing more bundles…**

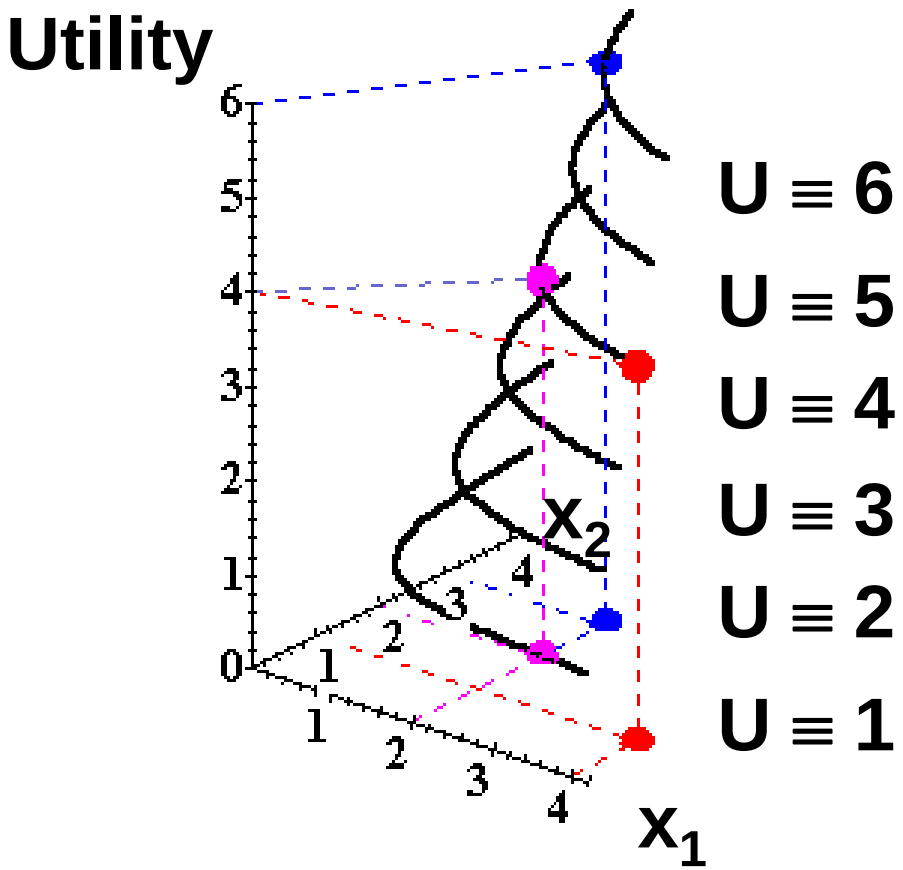
Utility Functions & Indiff. Curves



Utility Functions & Indiff. Curves

- **As before, this can be visualized in 3D by plotting each indifference curve at the height of its utility index.**

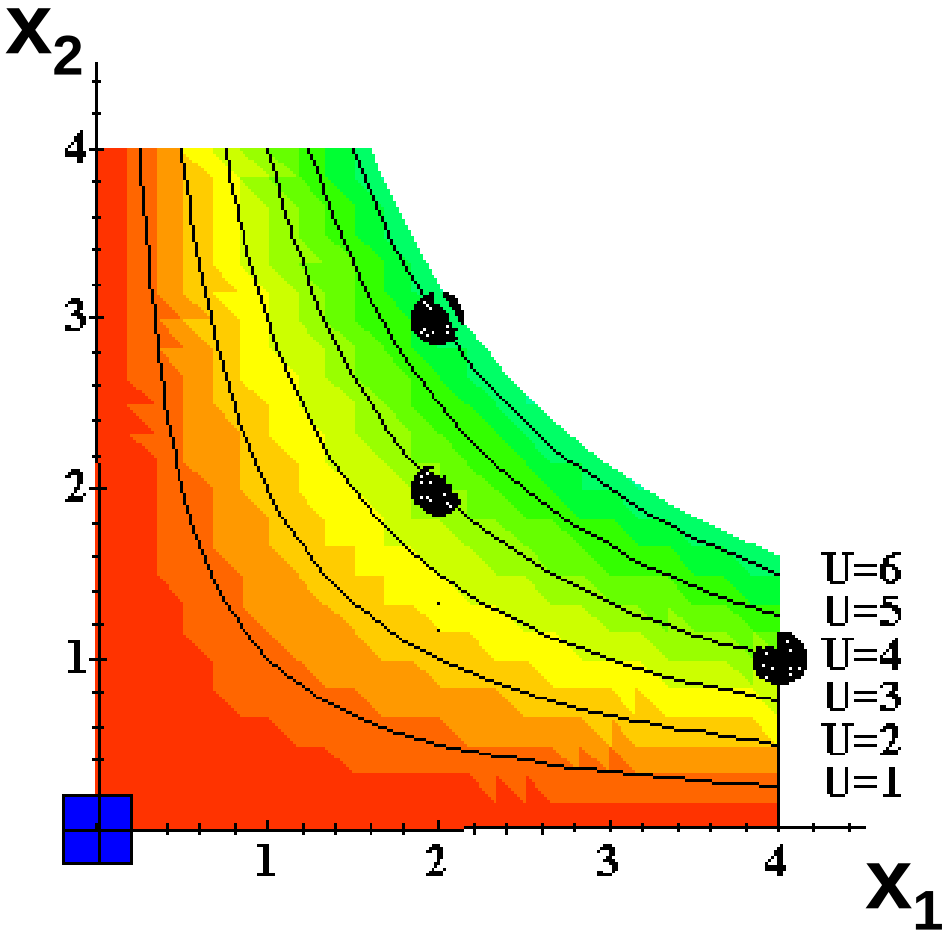
Utility Functions & Indiff. Curves



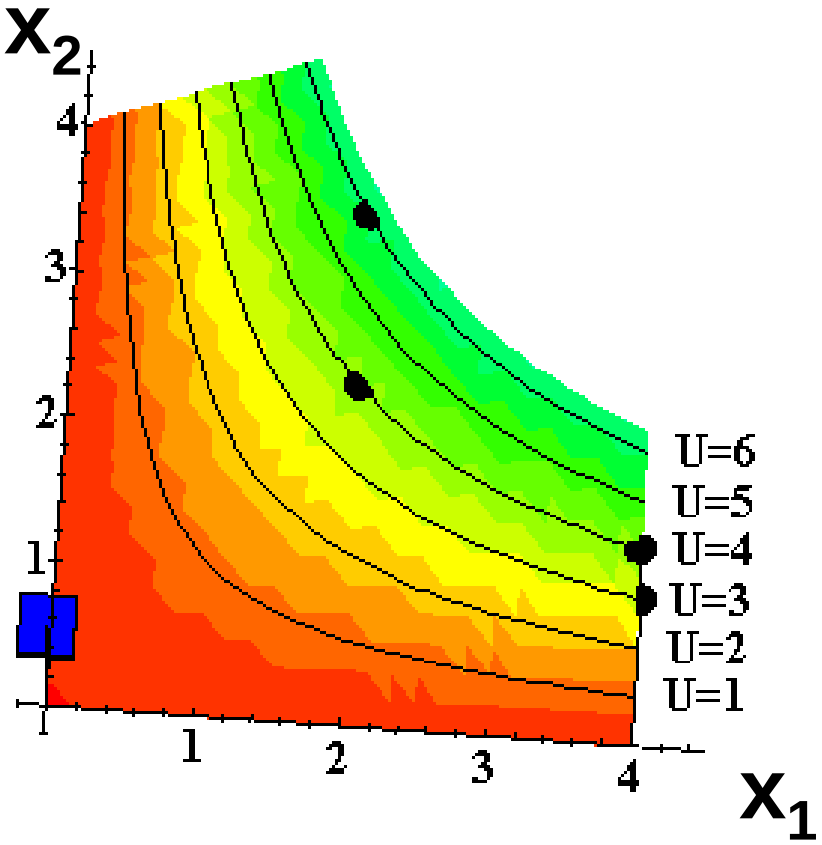
Utility Functions & Indiff. Curves

- **The whole collection of indifference curves completely represents the consumer's preferences.**

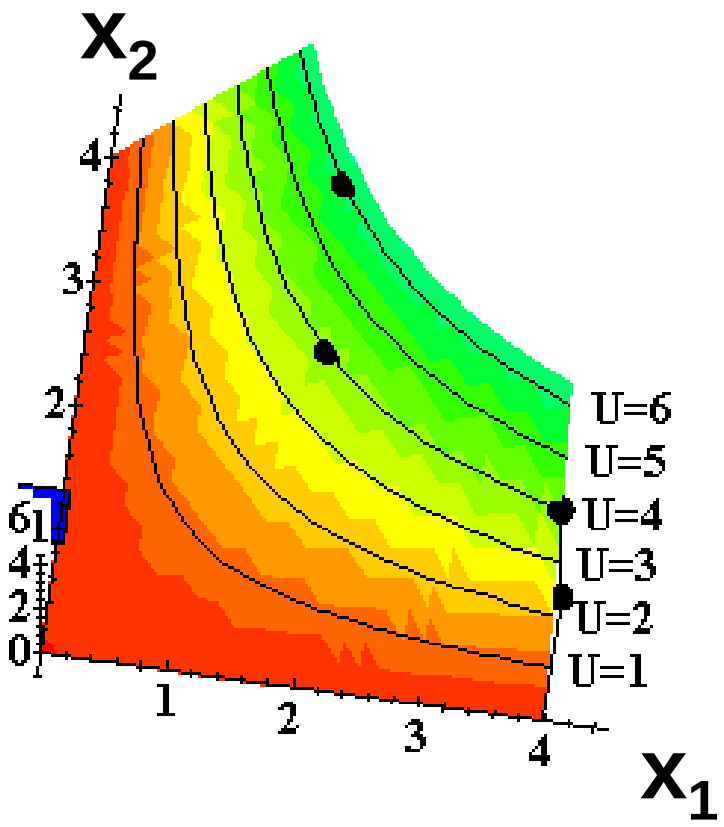
Utility Functions & Indiff. Curves



Utility Functions & Indiff. Curves



Utility Functions & Indiff. Curves



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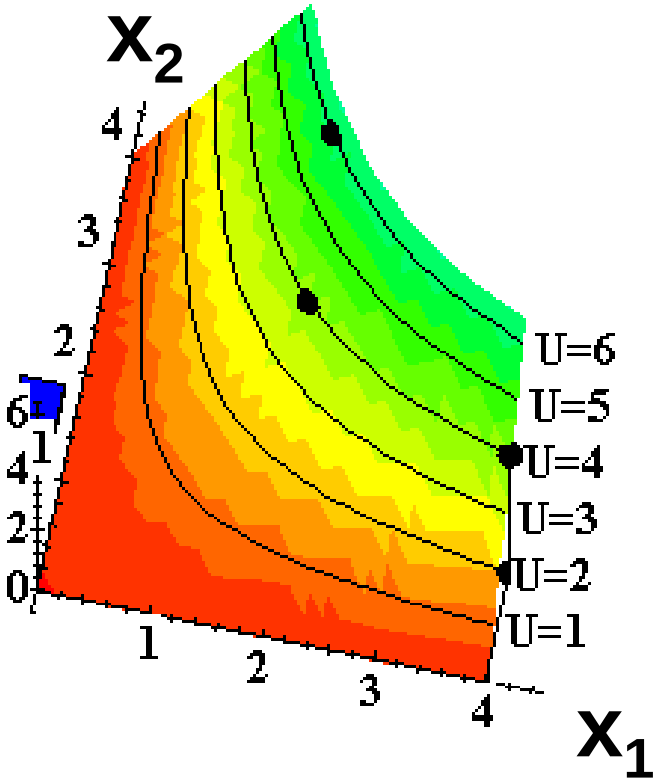

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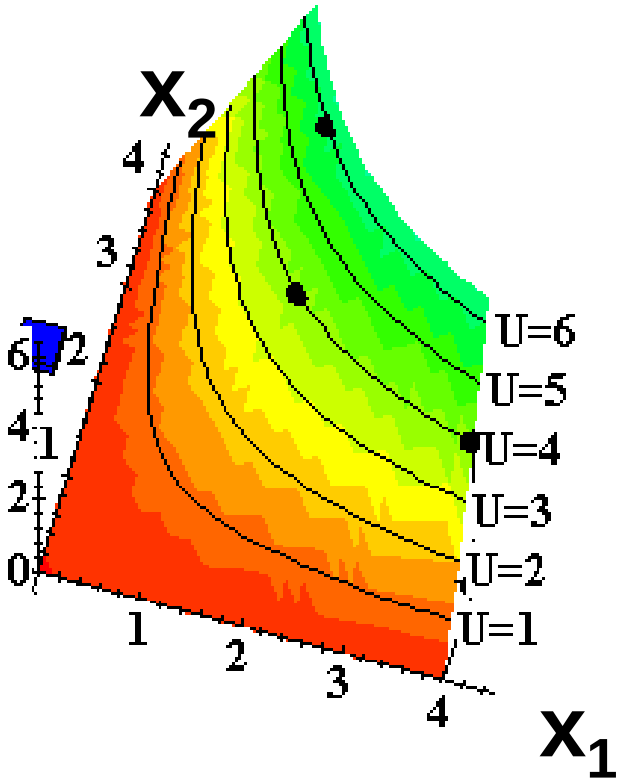

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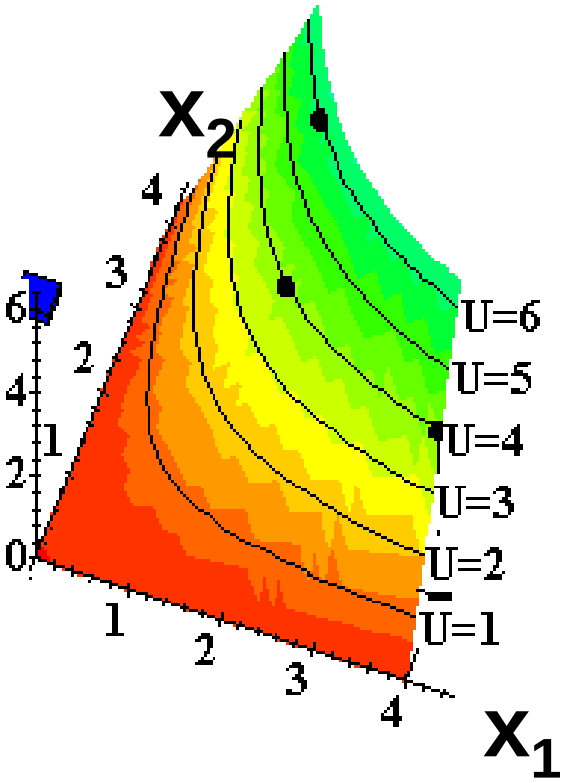

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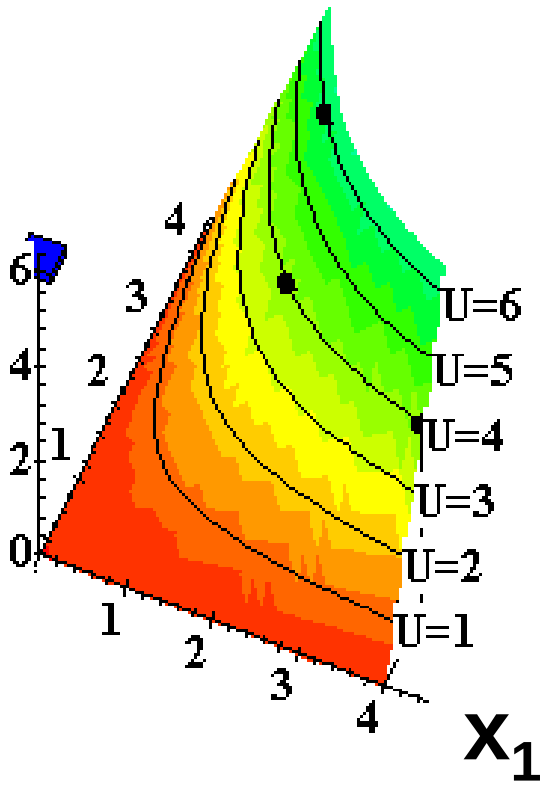

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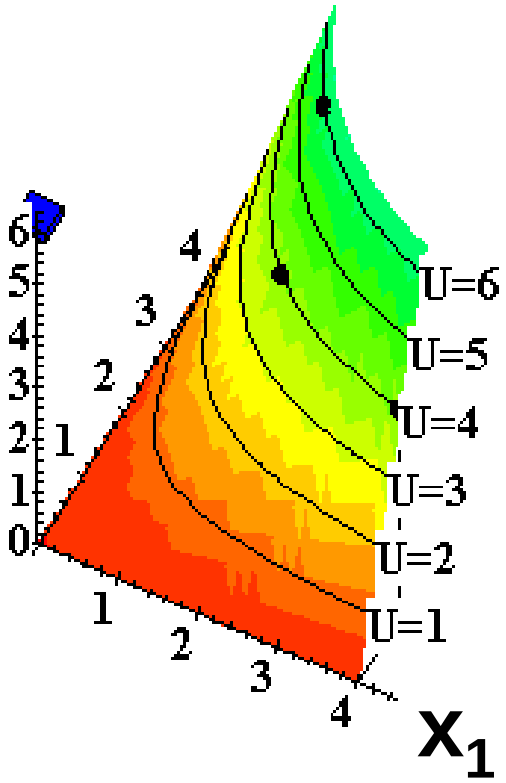

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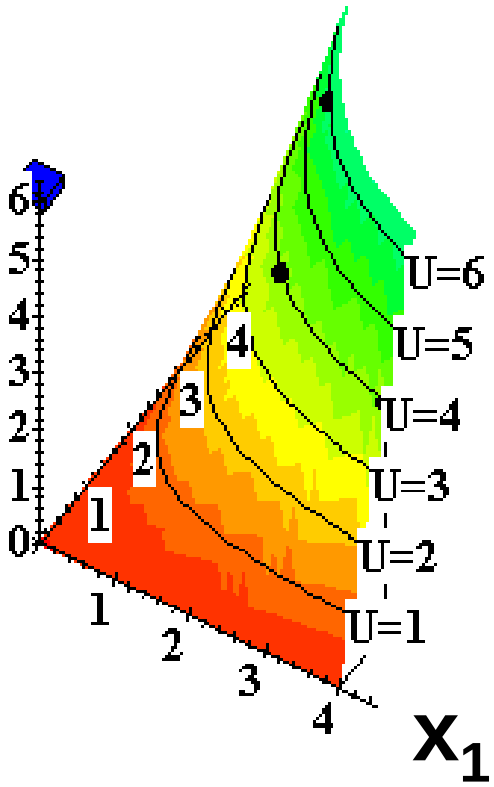

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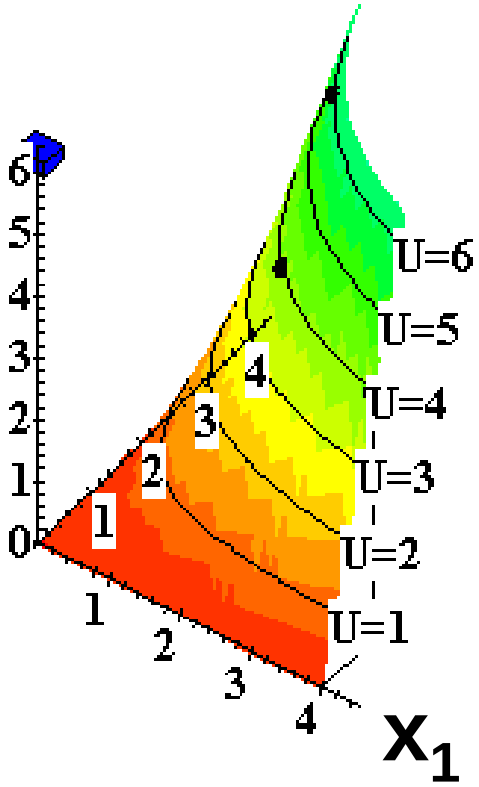

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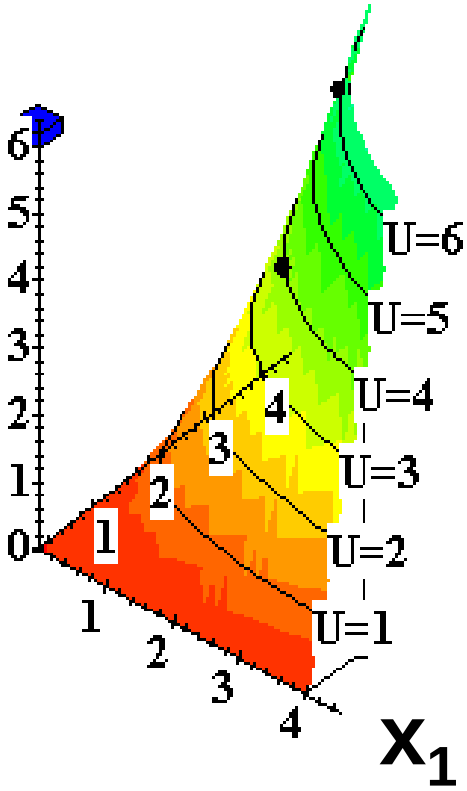

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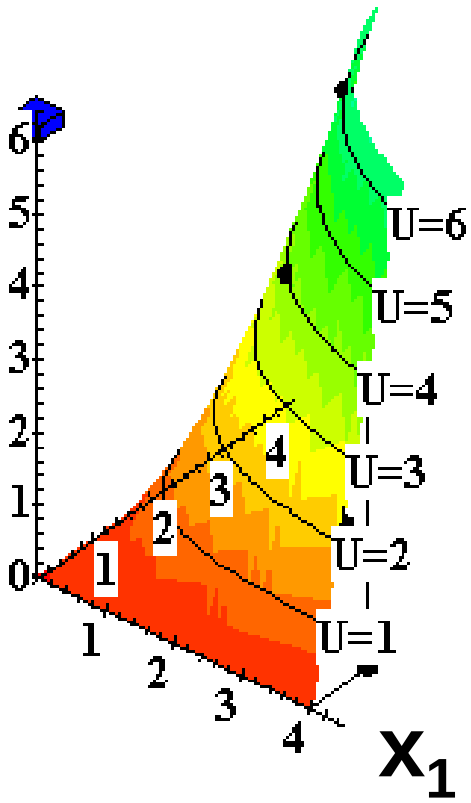

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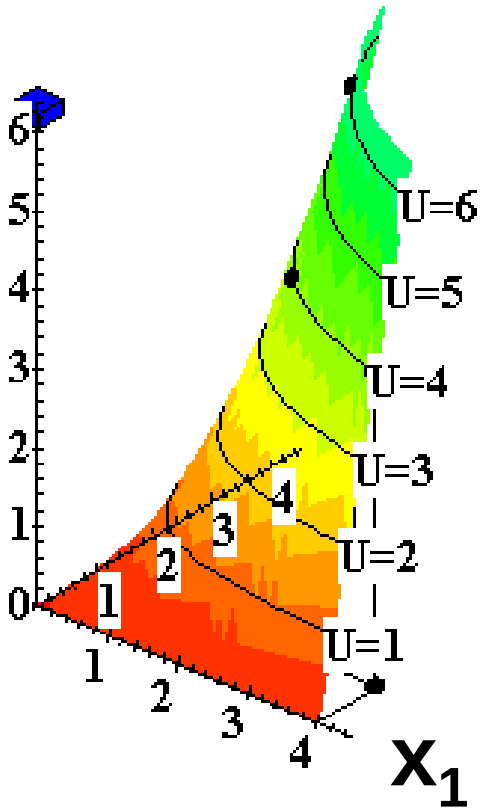

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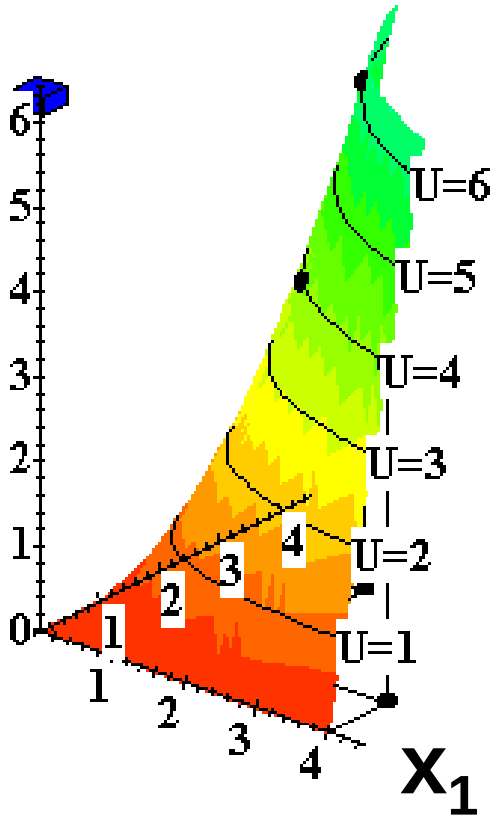

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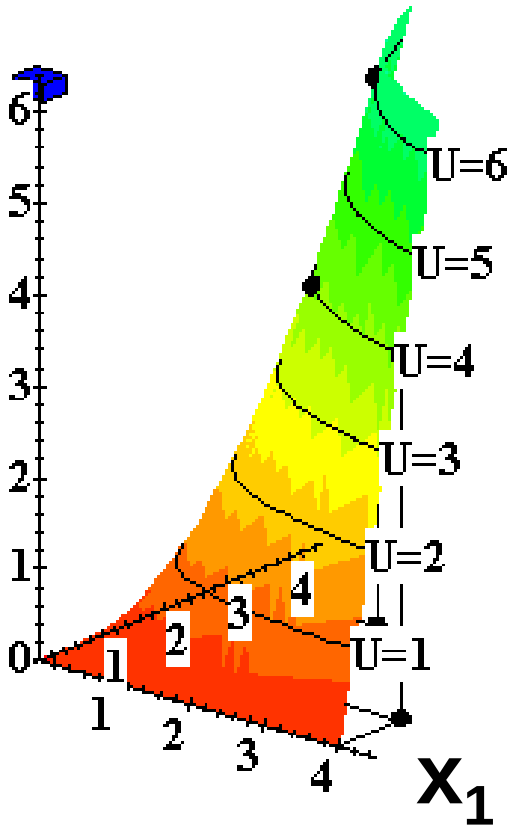

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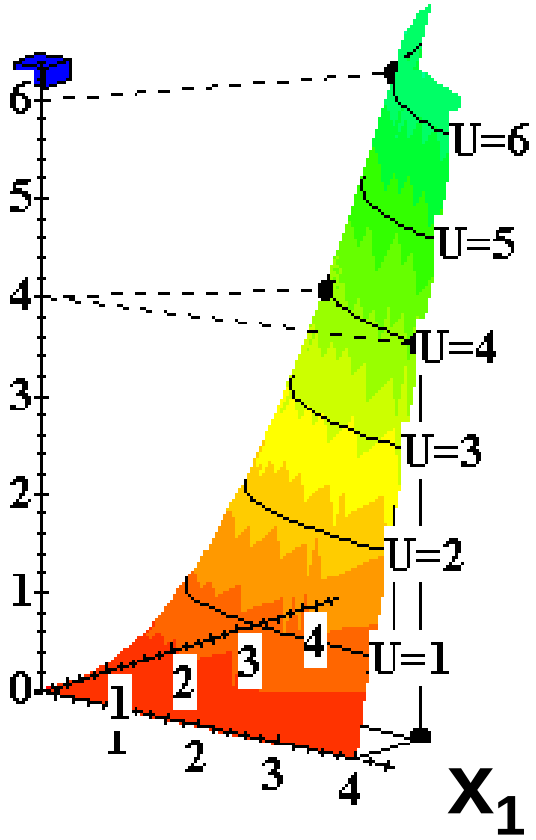


Utility Functions

- **No unique representation of pref**
- **$U(x_1,x_2) = x_1x_2 \Rightarrow (2,3) \succ (4,1) \sim (2,2)$.**
- **Define $V = U^2$. Then $V(x_1,x_2) = x_1^2x_2^2$**
- **$V(2,3) = 36 > V(4,1) = V(2,2) = 16$ so again $(2,3) \succ (4,1) \sim (2,2)$.**
- **V represents the same preferences.**

Utility Functions

- $U(x_1,x_2) = x_1x_2 \Rightarrow (2,3) \succ (4,1) \sim (2,2)$.
- Define $W = 2U + 10$.

Utility Functions

- $U(x_1,x_2) = x_1x_2 \Rightarrow (2,3) \succ (4,1) \sim (2,2)$.
- Define $W = 2U + 10$.
- Then $W(x_1,x_2) = 2x_1x_2+10$ so
  $W(2,3) = 22 > W(4,1) = W(2,2) = 18$.
  Again,
  $(2,3) \succ (4,1) \sim (2,2)$.
- W represents the same preferences.

Utility Functions

- **If**
  - **U is a utility function that represents a preference relation $\succsim$ and**
  - **f is a strictly increasing function,**
- **then V = f(U) is also a utility function representing $\succsim$.**

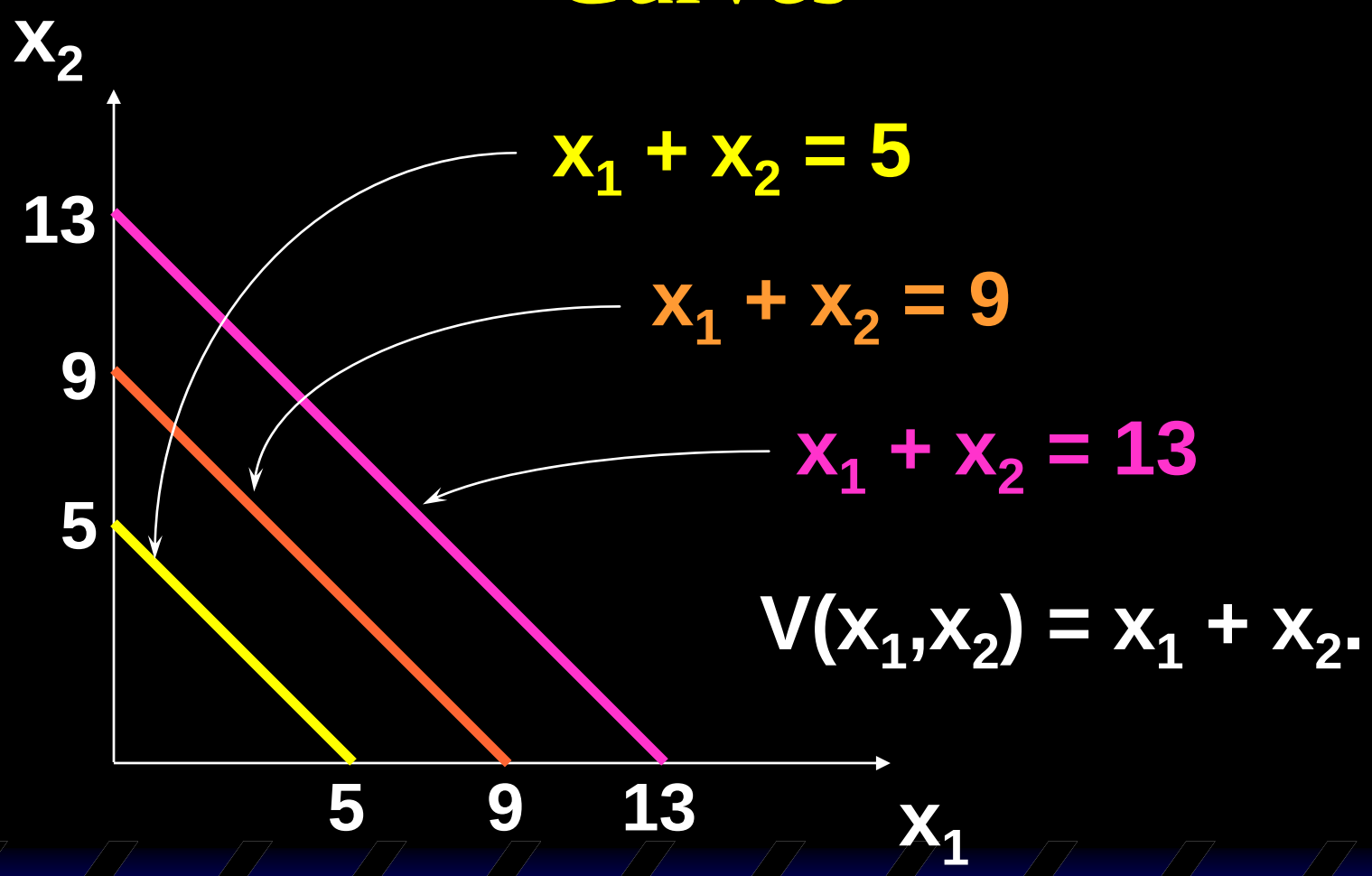
Some Other Utility Functions and Their Indifference Curves

- **Consider**

$$V(x_1,x_2) = x_1 + x_2.$$

What do the indifference curves for this utility function look like?

Perfect Substitution Indifference Curves



$V(x_1,x_2) = x_1 + x_2$.

All are linear and parallel.

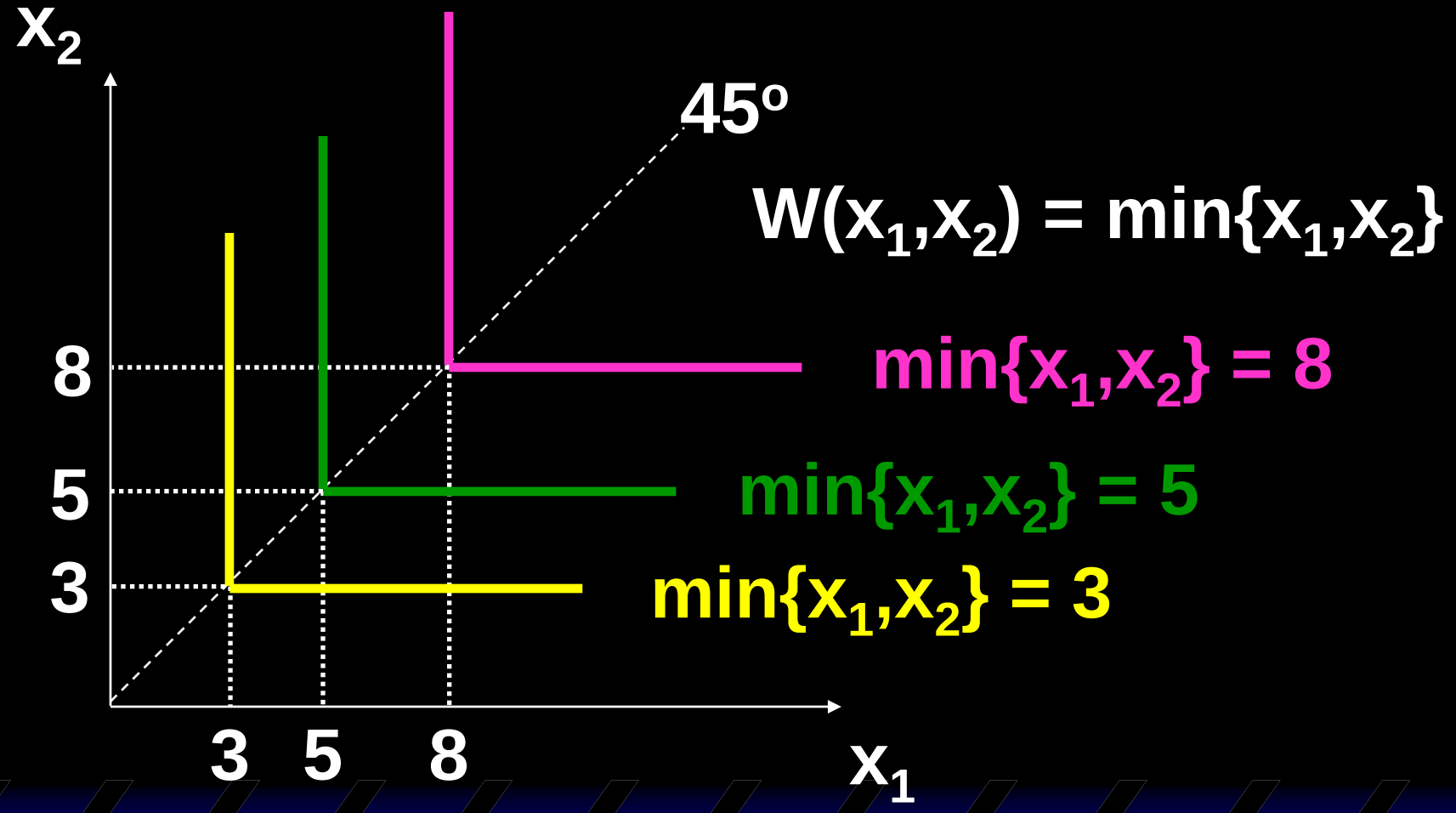
Some Other Utility Functions and Their Indifference Curves

- **Consider**

$$W(x_1,x_2) = \min\{x_1,x_2\}.$$

What do the indifference curves for this utility function look like?

Perfect Complementarity Indifference Curves



Some Other Utility Functions and Their Indifference Curves

- **A utility function of the form**

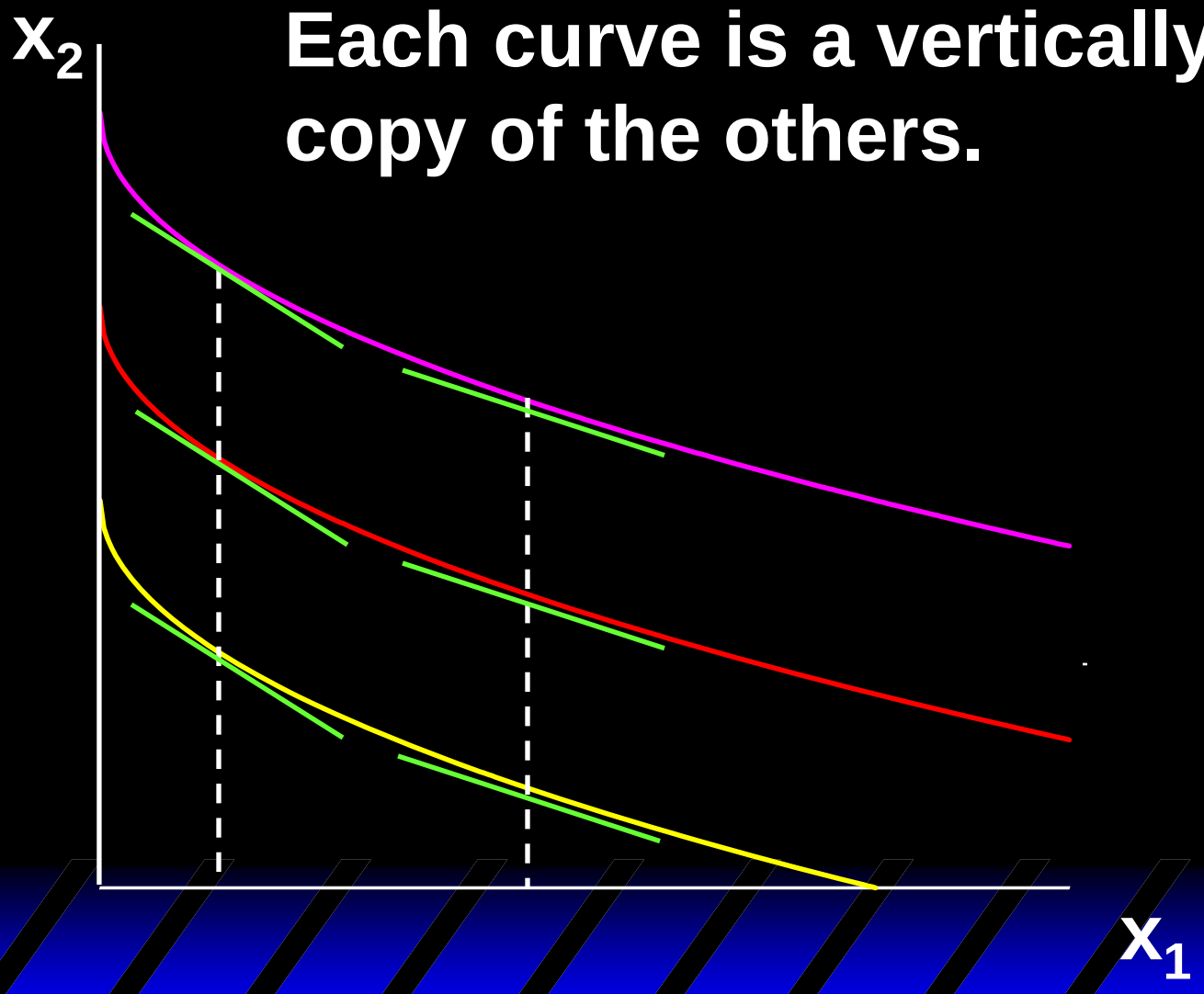
$$U(x_1,x_2) = f(x_1) + x_2$$

is linear in just x_2 and is called quasi-linear.

- ***E.g.*** $U(x_1,x_2) = 2x_1^{1/2} + x_2$.

Quasi-linear Indifference Curves

Each curve is a vertically shifted copy of the others.



Some Other Utility Functions and Their Indifference Curves

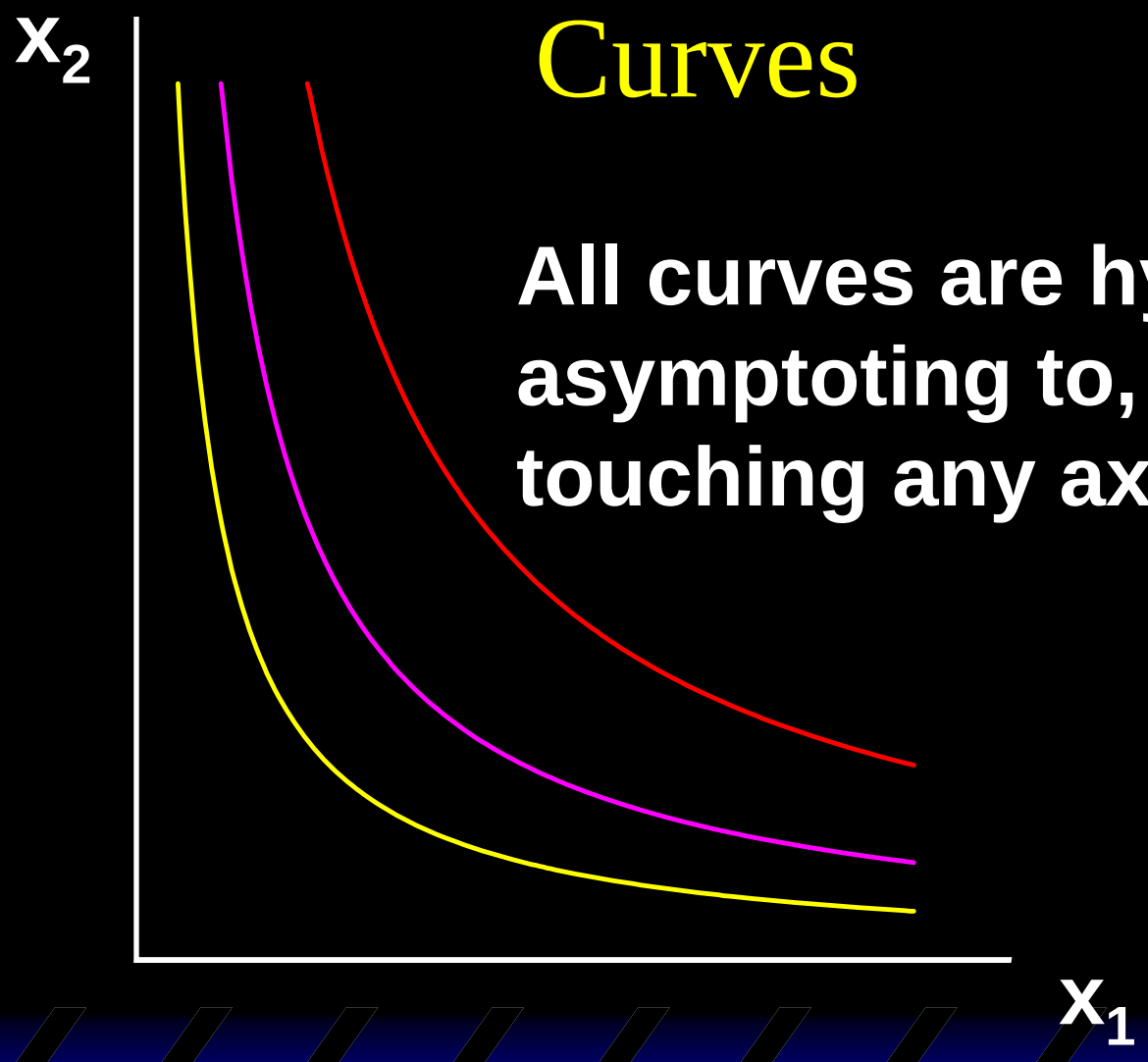
- **Any utility function of the form**

$$U(x_1,x_2) = x_1^a x_2^b$$

with a > 0 and b > 0 is called a Cobb-Douglas utility function.

- ***E.g.*** $U(x_1,x_2) = x_1^{1/2} x_2^{1/2}$ **(a = b = 1/2)**
  $V(x_1,x_2) = x_1 x_2^3$ **(a = 1, b = 3)**

Cobb-Douglas Indifference Curves



All curves are hyperbolic, asymptoting to, but never touching any axis.

Marginal Utilities

- **Marginal means "incremental".**
- **The marginal utility of commodity i is the incremental change of total utility that the consumer receives from a small change in the quantity of i**

$$MU_i = \frac{\partial U}{\partial x_i}$$

Marginal Utilities

- ***So*, if $U(x_1,x_2) = x_1^{1/2} x_2^2$ then**

$$MU_1 = \frac{\partial U}{\partial x_1} = \frac{1}{2} x_1^{-1/2} x_2^2$$

$$MU_2 = \frac{\partial U}{\partial x_2} = 2 x_1^{1/2} x_2$$

Marginal Utilities and Marginal Rates-of-Substitution

- **The general equation for an indifference curve is**

  $U(x_1,x_2) \equiv k$, a constant.

  Totally differentiating this identity gives

$$\frac{\partial U}{\partial x_1} dx_1 + \frac{\partial U}{\partial x_2} dx_2 = 0$$

Marginal Utilities and Marginal Rates-of-Substitution

$$\frac{\partial U}{\partial x_1} dx_1 + \frac{\partial U}{\partial x_2} dx_2 = 0$$

rearranged is

$$\frac{\partial U}{\partial x_2} dx_2 = -\frac{\partial U}{\partial x_1} dx_1$$

Marginal Utilities and Marginal Rates-of-Substitution

And $$\frac{\partial U}{\partial x_2} dx_2 = -\frac{\partial U}{\partial x_1} dx_1$$

rearranged is

$$\frac{dx_2}{dx_1} = -\frac{\partial U / \partial x_1}{\partial U / \partial x_2}.$$

This is the MRS.

Marg. Utilities & Marg. Rates-of-Substitution; An example

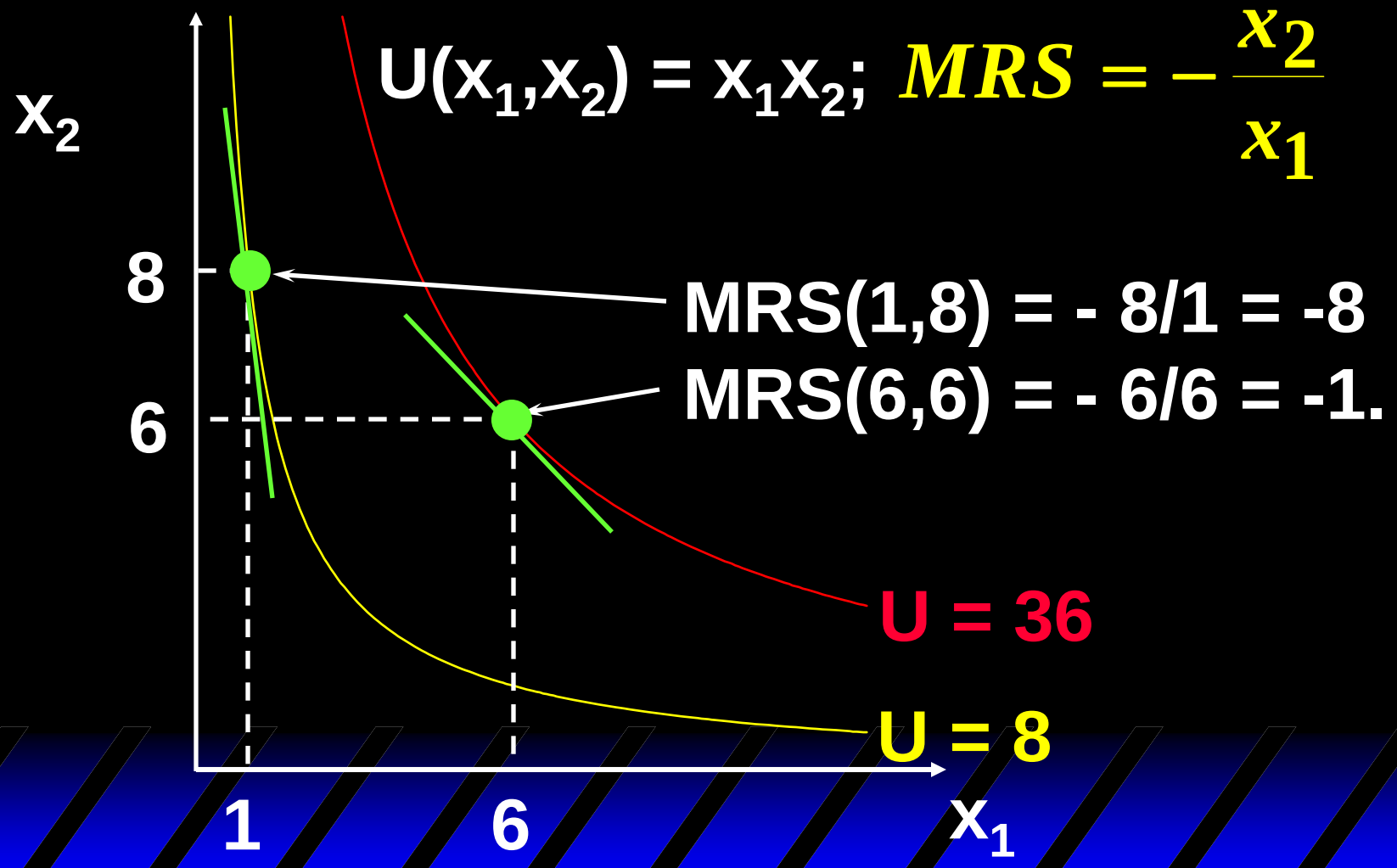
- **Suppose $U(x_1,x_2) = x_1x_2$. Then**

$$\frac{\partial U}{\partial x_1} = (1)(x_2) = x_2$$

$$\frac{\partial U}{\partial x_2} = (x_1)(1) = x_1$$

so $$MRS = \frac{dx_2}{dx_1} = -\frac{\partial U / \partial x_1}{\partial U / \partial x_2} = -\frac{x_2}{x_1}.$$

Marg. Utilities & Marg. Rates-of-Substitution; An example



Monotonic Transformations & Marginal Rates-of-Substitution

- **Applying a monotonic transformation to a utility function representing a preference relation simply creates another utility function representing the same preference relation.**
- **What happens to marginal rates-of-substitution when a monotonic transformation is applied?**

Monotonic Transformations & Marginal Rates-of-Substitution

- **More generally, if V = f(U) where f is a strictly increasing function, then**

$$MRS = -\frac{\partial V / \partial x_1}{\partial V / \partial x_2} = -\frac{f'(U) \times \partial U / \partial x_1}{f'(U) \times \partial U / \partial x_2}$$

$$= -\frac{\partial U / \partial x_1}{\partial U / \partial x_2}.$$

So MRS is unchanged by a positive monotonic transformation.